

ZIQIAN WANG

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EDUCATION

University of Michigan, Ann Arbor

Fall 2026

Bachelor of Science and Engineering in Computer Science

Shanghai Jiao Tong University

Sep 2024 – Present

Bachelor of Engineering in Electrical and Computer Engineering

RESEARCH EXPERIENCE

AgiBot, Shanghai

May 2026 – Present

Research Intern

Shanghai, China

- Built an end-to-end **SFT and RL post-training framework** for **flow-matching VLA policies (Pi0.5)**, integrating real-robot deployment, rollout data collection and processing, and distributed training into a closed loop.
- Implemented and benchmarked state-of-the-art **RL post-training algorithms** within this framework, including **DS-RL, RLT, and Q-VGM**.
- Applied these algorithms to two challenging real-robot tasks, raising success rates from **75% to 95%** on **cloth folding** (deformable, long-horizon) and from **33% to 79%** on **phone-charger plug insertion** (high-precision, contact-rich).

ScaleLab, Shanghai Jiao Tong University

Apr 2025 – Present

Research Intern (Advisors: Prof. Yao Mu, Prof. Qin Tong)

Shanghai, China

- “**Q-VGM**” (first author, arXiv preprint): proposed an **offline-to-online RL** framework for fine-tuning flow-matching VLAs (e.g., **Pi0.5**) that converts critic action-gradients into **velocity-field supervision**, bypassing the unstable **BPTT** through the denoising chain required by classical actor-critic updates.
- Improved the few-shot-SFT policy from **86.9% to 93.0%** on **LIBERO** with only 150 rollout episodes per task (offline), and to **99.7%** with offline-to-online training, matching PPO’s final performance with **~6× fewer rollout episodes**; on three **real-robot bimanual manipulation** tasks, raised average success from **66.7% to 98.3%**.
- “**Scene2Gym**” (second author, accepted at **IROS 2026**): co-developed a framework that converts real-world scenes into interactive, metric-consistent **3DGS** digital twins for embodied policy learning.
- Led the **RL and imitation learning policy training** pipeline within Scene2Gym, developing and tuning PPO-based and behavior-cloning policies for contact-rich manipulation tasks; achieved **56%** zero-shot **sim-to-real transfer** for visual RL policies and near-demonstration-level performance for imitation learning.

PUBLICATIONS

Z. Wang, et al., “Q-VGM: Q-Guided Value-Gradient Matching for Offline-to-Online RL of Flow-Matching VLA,” *Under Review*. (**First Author**) [arXiv]

Author, **Z. Wang** et al., “Scene2Gym: Transforming Real-World Scenes into Interactive Training Environments for Robot Learning,” *Accepted at IROS 2026*. (Second Author)

RESEARCH INTERESTS

I am interested in building embodied agents that interact with the physical world. Specifically, I aim to explore how reinforcement learning, foundation models, and world models can be combined to build generalist robots.

TECHNICAL SKILLS

Programming
Frameworks

Python, C++ (NOIP First Prize), MATLAB
PyTorch, MuJoCo, Isaac Gym, ROS, Weights & Biases, Claude Code